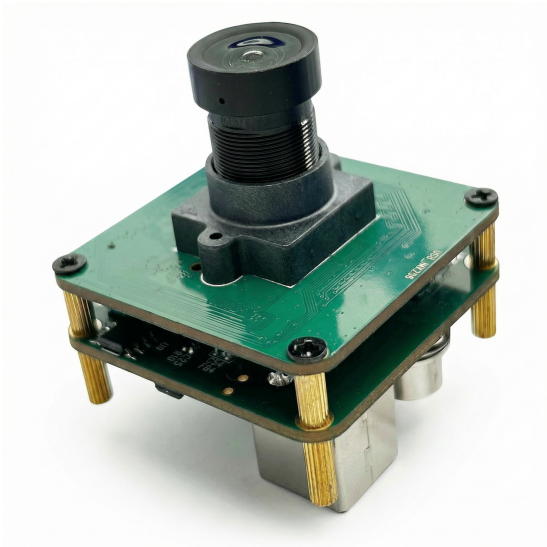


U3V-CAM-IMX296

USB3 Vision Industrial Camera

User Manual



U3V-CAM-IMX296 USB3 Vision Industrial Camera

Sensor	Sony IMX296LLR (Monochrome, Global Shutter)
Resolution	1.58 MP 1456 × 1088
Frame Rate	60 fps @ Full ROI
Interface	USB 3.0 USB3 Vision v1.0 / GenICam 3.x
Power	USB Bus-powered, 5V / ≤ 3.2W
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1. Product Overview

The **U3V-CAM-IMX296** is a high-performance USB3 Vision industrial camera featuring the **Sony IMX296LLR** monochrome global shutter CMOS sensor. With a resolution of **1.58 MP (1456 × 1088)** and a full-resolution frame rate of **60 fps**, it provides reliable, distortion-free imaging for demanding machine-vision applications including motion analysis, automation, robotics, and scientific imaging.

The camera is 100% compliant with **USB3 Vision v1.0** and **GenICam 3.x** standards, offering plug-and-play compatibility across all major software platforms. The InnoMaker SDK provides a lightweight, cross-platform C/C++ API and Python binding for direct, low-level camera access and maximum performance.

1.1 Key Features

Feature	Description
Sony IMX296 Global Shutter	1/2.9" CMOS sensor, 3.45 μm pixels — zero rolling distortion
High Frame Rate	60 fps at full 1456×1088; exceeds 1900 fps with reduced ROI
True Analog Gain	0–48 dB range, 0.1 dB steps — extremely low-noise imaging
Long Exposure	Manual shutter with true long-exposure capability (≥ 15 seconds)
Precise Synchronization	Opto-isolated hardware trigger (5–24V) + strobe output, 0.1 μs timestamping
Industrial Design	USB 3.0 bus-powered, compact, on-board temperature monitoring
Standard Compliance	Fully compatible with any GenICam-compliant software
InnoMaker SDK	Cross-platform C/C++ + Python SDK — no proprietary patches required

1.2 Specifications

Parameter	Specification
Sensor	Sony IMX296LLR (Monochrome, Global Shutter)
Resolution	1456 (H) × 1088 (V), 1.58 MP
Pixel Size	3.45 μm × 3.45 μm
Optical Format	1/2.9"
Max Frame Rate	60 fps @ Full ROI (Mono8)
Exposure Time	29 μs to ≥ 15 seconds
Analog Gain	0 – 48 dB (0.1 dB steps)
Interface	USB 3.0 (Type-C or Micro-B depending on model)
I/O Connector	6-pin Hirose HR10A-7R-6P (Trigger In, Strobe Out, GND)
Power	USB Bus-powered, 5V / ≤ 3.2W
Lens Mount	M12 (Default) / CS-Mount (Optional)
Operating Temperature	-10 °C to +65 °C
Standards	USB3 Vision v1.0, GenICam 3.x

1.3 Industry Applications

Application	Use Case
Machine Vision & Automation	AOI inspection, robotics, quality control, pick-and-place
Motion Analysis	High-speed capture, sports science, ballistics
Scientific Imaging	Microscopy, fluorescence, diagnostics
Security & Surveillance	24/7 monitoring, license plate recognition
Broadcast & ProAV	Live streaming, drone cameras, cinema

2. Hardware

2.1 6-pin Hirose Connector Pinout (HR10A-7R-6P)

The camera exposes a 6-pin Hirose connector for hardware trigger input and strobe output. Both trigger and strobe lines are opto-isolated for noise immunity in industrial environments.

Pin	Signal	Description
1	GPIO_B33	Reserved
2	Trig +	Opto-isolated Trigger Input (+)
3	GPIO_A33	Reserved
4	STROBE +	Opto-isolated Strobe Output (+)
5	STROBE – / Trig –	Opto-isolated I/O Ground
6	GND	System Ground

Note: Trigger input voltage range: 5–24V (opto-isolated). For detailed hardware dimensions and wiring diagrams, refer to the U3V-CAM-IMX296 User Manual V10.pdf included in the repository.

3. InnoMaker SDK

The **InnoMaker U3V SDK** (version 2.1.0) is the recommended choice for developers who need direct, low-level camera access and maximum performance. It provides a lightweight, cross-platform C/C++ API and a Python binding — all built on the same underlying shared library. Code written against the C API on Windows compiles and runs unchanged on Linux; Python code runs unchanged on every supported platform.

3.1 SDK Packages Overview

Package	Platform	Size	Description
V9-SDK-DLL-CUS.zip	Windows x64	26 MB (zip) 56 MB extracted	C/C++ SDK with GUI viewer, CLI demo, headers, bundled Qt6
V9-SDK-SO-CUS.tar.gz	Linux x64 / ARM64	520 KB (tar.gz) 1.4 MB extracted	C/C++ SDK with GUI viewer, CLI demo, and headers
V9-SDK-PYTHON-CUS.zip	Windows + Linux x64 / ARM64	540 KB (zip) 1.2 MB extracted	Python SDK — bundles native libraries for all platforms; ideal

All three packages share the same SDK source tree and the same underlying shared library. The Python package wraps the identical C library via ctypes — there is no separate codepath.

3.2 Supported Platforms

Platform	Architecture	Status
Windows 10 / 11	x64	Fully Supported
Ubuntu 22.04+ / Debian 12+	x64	Fully Supported
Raspberry Pi 5 (Bookworm / Trixie)	ARM64	Fully Supported
NVIDIA Jetson Orin Nano (JetPack 6.0+)	ARM64	Fully Supported

3.3 Windows Installation (C/C++)

The Windows package is fully self-contained — Qt6, Visual Studio runtime, libusb, and the USB driver installer are all bundled. No prerequisites are required.

```
1. Extract V9-SDK-DLL-CUS.zip to any directory (e.g., D:\u3v\)  
2. Plug in the U3V camera  
3. Right-click V9-SDK-DLL-CUS\tools\zadig-2.9.exe  
→ Run as administrator  
→ Follow WINUSB_DRIVER_INSTALL.md to install the WinUSB driver  
4. Double-click V9-SDK-DLL-CUS\run_viewer.bat  
→ GUI launches; camera appears in device list
```

For application development, add the following to your MSVC project:

```
#include <u3v/u3v_sdk.h>  
  
// MSVC project settings:  
// Additional Include Directories: V9-SDK-DLL-CUS\include  
// Additional Library Directories: V9-SDK-DLL-CUS\lib  
// Additional Dependencies: u3v_cam.lib  
// At runtime, ensure V9-SDK-DLL-CUS\bin is on PATH
```

3.4 Linux Installation (C/C++)

The Linux package supports both x86_64 and ARM64 (Raspberry Pi 5, NVIDIA Jetson, Rockchip SBCs). Only two system packages are required: **libusb-1.0-0** and **libqt6widgets6**.

```
# 1. Extract
tar xzf V9-SDK-S0-CUS.tar.gz
cd V9-SDK-S0-CUS

# 2. Install runtime dependencies (one-time)
sudo apt update && sudo apt install -y libusb-1.0-0 libqt6widgets6

# 3. Install udev rule (allow non-root USB camera access, one-time)
sudo tee /etc/udev/rules.d/99-u3v.rules > /dev/null <<'EOF'
SUBSYSTEM=="usb", ATTRS{bDeviceClass}=="ef", ATTRS{bDeviceSubClass}=="02", \
ATTRS{bDeviceProtocol}=="01", MODE="0666", GROUP="plugdev"
EOF
sudo udevadm control --reload-rules && sudo udevadm trigger
sudo usermod -aG plugdev $USER
# Log out and back in for the group change to take effect

# 4. Launch the GUI viewer
case "$(uname -m)" in
x86_64) cd ubuntu22.04-x64 ;;
aarch64) cd ubuntu22.04-arm64 ;;
esac
./run_viewer.sh
```

For application development (GCC):

```
ARCH_DIR=ubuntu22.04-x64 # or ubuntu22.04-arm64

gcc my_app.c \
-I$ARCH_DIR/include \
-L$ARCH_DIR/lib -lu3v_cam \
-Wl, -rpath,'$ORIGIN/lib' \
-o my_app
```

3.5 Python SDK (V9-SDK-PYTHON-CUS)

The Python SDK targets **Python 3.8–3.12** on Windows x64, Linux x64, and Linux ARM64. It is ideal for ML / computer-vision workflows, rapid prototyping, headless capture scripts, and Jupyter notebooks. The package wraps the same C library via ctypes — no separate codepath.

Windows:

```
:: 1. Extract V9-SDK-PYTHON-CUS.zip
:: 2. Install winUSB driver (use Zadig from V9-SDK-DLL-CUS\tools\
:: 3. Open a command prompt in V9-SDK-PYTHON-CUS\
install_deps.bat :: installs numpy + optional PyQt6 / opencv
run_basic_capture.bat :: smoke test – captures 5 frames
run_viewer.bat :: live PyQt6 viewer
```

Linux (Ubuntu 22.04, Debian 12+, Raspberry Pi OS Bookworm):

```
# Same prerequisites as C/C++ Linux package (libusb-1.0-0 + udev rule)
unzip V9-SDK-PYTHON-CUS.zip
cd V9-SDK-PYTHON-CUS
chmod +x *.sh
./install_deps.sh
./run_basic_capture.sh
./run_viewer.sh
```

Python API Example:

```
import u3v_cam

# Discover cameras
print(u3v_cam.list_cameras())

# Open, configure, capture
with u3v_cam.Camera() as cam:
    cam.set_roi(1456, 1088)
    cam.pixel_format = u3v_cam.PFNC_MONO8
    cam.exposure_us = 5000 # 5 ms
    cam.gain = 0
    cam.frame_rate = 60
    cam.start()
    for _ in range(60):
        frame = cam.read_frame() # numpy.ndarray (H, W) uint8
    cam.stop()
```

Optional Dependencies:

Feature	Package	Install Command
Core (required)	numpy >= 1.20	pip install numpy
PyQt6 live viewer	PyQt6 + pyqtgraph	pip install PyQt6 pyqtgraph
Live preview	opencv-python	pip install opencv-python

3.6 C API Reference

The SDK exposes a clean C API of 74 functions. The key functions for initialization, discovery, camera control, and streaming are listed below. For the complete reference, see the headers in **include/u3v/**.

```
/* Initialization */
u3v_status_t u3v_sdk_init(void);
void u3v_sdk_shutdown(void);

/* Discovery */
int u3v_discover(u3v_device_info_t *info, int max_devices);

/* Camera control */
u3v_status_t u3v_camera_open(u3v_camera_t **cam, int device_index);
void u3v_camera_close(u3v_camera_t *cam);
u3v_status_t u3v_camera_set_exposure(u3v_camera_t *cam, uint32_t microseconds);
u3v_status_t u3v_camera_set_gain(u3v_camera_t *cam, uint32_t gain);
u3v_status_t u3v_camera_set_trigger_mode(u3v_camera_t *cam, uint32_t mode);

/* Streaming */
u3v_status_t u3v_stream_create(u3v_stream_t **stream, u3v_camera_t *cam,
const u3v_stream_config_t *config);
void u3v_stream_destroy(u3v_stream_t *stream);
u3v_status_t u3v_stream_grab(u3v_stream_t *stream, u3v_frame_t *frame,
uint32_t timeout_ms);
```

Header File	Contents
u3v_sdk.h	Core SDK init / shutdown
u3v_camera.h	Camera open / control / configuration
u3v_stream.h	Streaming setup and frame grab
u3v_types.h	Common types and status codes
u3v_protocol.h	GenCP protocol primitives
u3v_imx296.h	Sensor-specific helpers for IMX296

4. Third-Party Software

The U3V-CAM-IMX296 is 100% USB3 Vision / GenICam compliant and works out-of-the-box with any standard-compliant software. The following third-party tools are verified and supported for users who prefer GUI-based or open-source GenICam workflows.

Tool	Platform	License	Notes
eBUS Player (Pleora / JAI)	Windows, Linux, Pi 5	Commercial (evaluation mode available)	Full featured GUI: live preview, parameter control, multi-camera support
Aravis	Linux	LGPL (free for commercial use)	Open-source GenICam implementation; lightweight alternative

Local packages for Raspberry Pi 5 are pre-downloaded in the repository under **eBusPlayer&Aravis_PI5_Linux/**. For Windows, download the latest eBUS SDK from the JAI website: <https://www.jai.com/support-software/jetson-ubuntu>

5. Resources & Downloads

Resource	Location / Link
InnoMaker SDK (Windows C/C++)	InnoMaker_SDK_Libusb_Win_Linux/V9-SDK-DLL-CUS.zip
InnoMaker SDK (Linux C/C++)	InnoMaker_SDK_Libusb_Win_Linux/V9-SDK-SO-CUS.tar.gz
InnoMaker SDK (Python)	InnoMaker_SDK_Libusb_Win_Linux/V9-SDK-PYTHON-CUS.zip
SDK Delivery Overview	InnoMaker_SDK_Libusb_Win_Linux/DELIVERY_OVERVIEW.md
eBUS Player (Raspberry Pi 5)	eBusPlayer&Aravis_PI5_Linux/ebus_for_raspberry_pi5/
Aravis (Raspberry Pi 5)	eBusPlayer&Aravis_PI5_Linux/aravis_for_raspberry_pi5/
Pre-installed OS Image (Pi 5)	https://www.jianguoyun.com/p/DXuEVqMQpdSrBxiqmp0GIAA (Password: uwpu3)
Hardware User Manual V10	U3V-CAM-IMX296 User Manual V10.pdf
GitHub Repository	https://github.com/INNO-MAKER/u3v-cam-imx296

6. Support

Channel	Details
Website	www.inno-maker.com
GitHub	github.com/INNO-MAKER
Technical Support	support@inno-maker.com
Sales	sales@inno-maker.com
Direct Contact	zoujy@inno-maker.com

For SDK-related issues, please include your OS, architecture, SDK version, and a minimal reproducible example when contacting support.